5.13.1 Constructing a parallelogram when its two adjacent sides and one angle are given

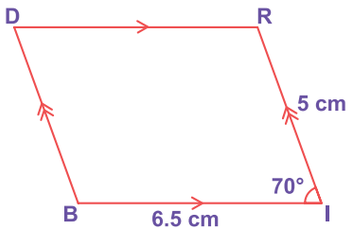
Example 5.30

Construct a parallelogram BIRD with BI=6.5*cm*, IR=5*cm* and $\angle BIR = 70°$. Also find its area.

Solution:

Given:

BI=6.5*cm*, IR=5*cm* and $\angle BIR = 70°$



Rough diagram

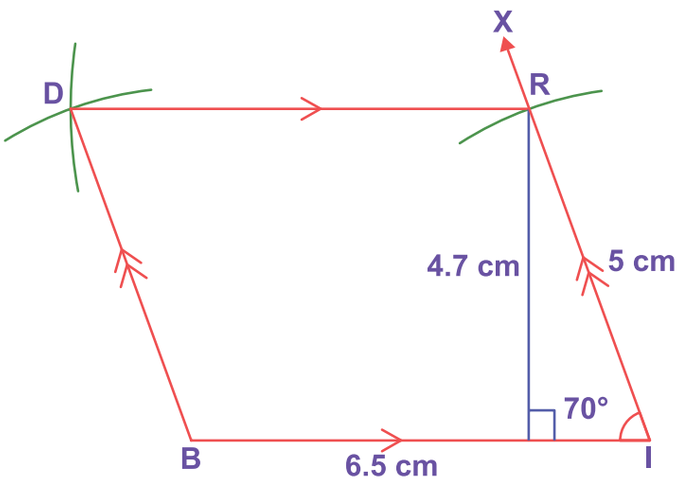


Fig. 5.49

Steps:

1. Draw a line segment BI=6.5*cm*.
2. Make an angle $\angle BIX = 70°$ at I on $\overline{BI}$.
3. With I as centre, draw an arc of radius 5*cm* cutting IX at R.
4. With B and R as centres, draw arcs of radii 5*cm* and 6.5*cm* respectively. Let them cut at D.
5. Join BD and RD.
6. BIRD is the required parallelogram.

Calculation of Area:

Area of the parallelogram BIRD $= bh$ sq.units

$= 6.5 \times 4.7 = 30.55\ sq.cm$

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